

NE 470/570 Monte Carlo Methods for Radiation Transport

Topic 5b: Fixed Source Monte Carlo Particle Transport

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Outline

1. Linear Boltzmann Equation
2. MC for Simplified Particle Transport
3. 1-D Monte Carlo algorithm
4. Statistical Reliability of MC Results

- This lecture is devoted to application of the Monte Carlo method to a **simplified particle transport problem**. Its associated **variance reduction** will be covered in Lecture 5c.
- The method provides the ability of performing experiments on a computer and estimating expected behavior of particles and their interactions in a medium.
- In general, a particle is emanated from a source (fixed or fission) with a **random spatial position**, **random direction**, and **random energy**. Each particle has a chance of traveling freely in a medium before undergoing an interaction with a nuclei.
- **Different types of interactions** may occur, depending on the particle type and energy, and the composition of the medium.
- These interactions, which can be described by PDFs that are established using nuclear data and physics models, **may lead to production of one or more particles, termination of the particle, and/or a change in particle energy or direction**.
- A particle may also be terminated if it escapes from the medium.
- In a MC simulation, the **"history"** of each particle from birth to death is followed, expected tallies of interest are estimated by simulation of numerous histories, and the associated variance and relative uncertainties for these tallies are evaluated.

Outline

1. Linear Boltzmann Equation
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- The **time-independent linear Boltzmann equation (LBE)** provides the particle balance in a phase space. In this topic we will focus on a **nonmultiplying medium**, but applications in multiplying media or eigenvalue problems are also important. The derivations in this topic are valid for any neutral particle transport, but in the course we will refer to **neutrons only**.
- The time-independent LBE for fixed-source, no fission source, without delayed neutrons as

$$\hat{\Omega} \cdot \nabla \psi(\mathbf{r}, E, \hat{\Omega}) + \Sigma_t(\mathbf{r}, E) \psi(\mathbf{r}, E, \hat{\Omega}) = \int_0^\infty dE' \int_{4\pi} d\hat{\Omega}' \Sigma_s(\mathbf{r}, E' \rightarrow E, \hat{\Omega} \cdot \hat{\Omega}') \psi(\mathbf{r}, E', \hat{\Omega}') + S(\mathbf{r}, E, \hat{\Omega}) \quad (1)$$

- $\psi(\mathbf{r}, E, \hat{\Omega}) = v(E)n(\mathbf{r}, E, \hat{\Omega})$ is the angular neutron flux in phase space, $v(E)$ is the particle speed at energy E , $n(\mathbf{r}, E, \hat{\Omega})$ is particle number density in phase space;
 - $\Sigma_t(\mathbf{r}, E)$ is the total macroscopic cross section indicating probability per unit length of particle-nuclei interaction of all types at position $\mathbf{r}$ and energy E ;
 - $\Sigma_s(\mathbf{r}, E' \rightarrow E, \hat{\Omega} \cdot \hat{\Omega}')$ is the differential macroscopic scattering cross-section, indicating probability per unit length that following a particle-nuclei scattering a particle will scatter from energy E' to energy E within dE , and from direction $\hat{\Omega}'$ to direction $\hat{\Omega}$ within $d\hat{\Omega}$;
 - $S(\mathbf{r}, E, \hat{\Omega})$ is the fixed source density.
- The MC method can be used by **averaging** histories (from birth to death) of many individual particles to calculate different terms in the LBE.

- Because the majority of MC concepts and techniques can be discussed without the need for multidimensionality and energy dependency, we will focus on **one-dimensional, one-speed particle transport, with isotropic elastic-scattering and source**, expressed by:

$$\mu \frac{\partial \psi(x, \mu)}{\partial x} + \Sigma_t \psi(x, \mu) = \frac{1}{2} (\Sigma_s \phi(x) + S(x, \mu)) \quad (2)$$

- x refers to the dimension in which physical properties are changing;
- μ refers to the **direction cosine** ($\mu = \cos \theta$) of the direction ($\hat{\Omega}$) relative to the x axis¹;
- $\psi(x, \mu)$ is the angular flux;
- $\phi(x)$ refers to the scalar flux obtained by integrating the angular flux, i.e.,

$$\phi(x) = \int_{-1}^1 \psi(x, \mu) d\mu;$$
- $S(x, \mu)$ is the source term. For further simplicity, the source is considered to be generated at a point (x_0) and along one direction (μ_n), $S(x, \mu) = S_0 \delta(x - x_0) \delta(\mu - \mu_n)$.

- We will only consider two types of interactions, **scattering** and **absorption**. This means

$$\Sigma_t = \Sigma_s + \Sigma_a \quad (3)$$

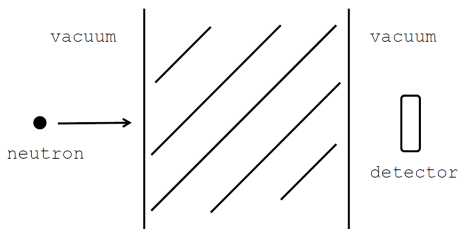
and absorption is considered same with **capture**, which terminates a particle history.

¹Note that in Lecture 5a, we defined the polar angle θ to be relative to the **z axis** in order to be consistent with the reference book by Prinja and Larsen. In this Equation, which is the same with the Equation (6.3) in the book by Haghghat, the direction cosine is w.r.t. the x -axis.

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- Consider a mono-directional, one-speed neutron point source placed at the left boundary of a 1-D **homogeneous** shield, which is placed in a **vacuum**, with a **detector** at its right boundary.
- In order to determine the neutron distribution within or beyond the shield, i.e., at the detector, or the rate of **transmission** or **reflection** of particles through the shield via the MC method, first we need to **identify the basic physical processes, and to obtain their associated PDFs**.
- Further, we need to derive the corresponding FVMCs for sampling random variables corresponding to different random processes. Note that since the shield is placed in a **vacuum**, any particle that leaves the shield will not return, i.e., **nonreentrant boundary condition**.
- In this simplified particle transport, there are only three basic random processes:
 - 1 **Path-length** (or free flight)
 - 2 **Interaction type** (absorption or scattering)
 - 3 **Scattering angle** in the case of scattering interaction (only “elastic” scattering is considered)



Outline

2. MC for Simplified Particle Transport

2.1. Sampling path length

2.2. Sampling interaction type

2.3. Selection of scattering angle

- We need to derive the combined probability of a particle moving freely a distance r , followed by probability of interaction in dr .
- From basic particle physics, we know that:
 - 1 probability of free travel in a distance r is $e^{-\Sigma_t r}$
 - 2 probability of an interaction in a distance dr is $\Sigma_t dr$
- Hence, the combined probability of above processes is given by

$$p(r) = \Sigma_t e^{-\Sigma_t r}, \quad 0 < r < +\infty \quad (4)$$

- The FFMC for this process is:

$$P(r) = \int_0^r \Sigma_t e^{-\Sigma_t r'} dr' = \eta \quad (5)$$

- The path length can be solved as:

$$1 - e^{-\Sigma_t r} = \eta \quad \rightarrow \quad \Sigma_t r = -\ln(1 - \eta) \quad \rightarrow \quad r = -\frac{\ln \eta}{\Sigma_t} \quad (6)$$

- Note that we have replace $(1 - \eta)$ with (η) , because they represent an equivalent sequence of random numbers.

- For a **multi-region problem**, Σ_t may change from one region to other, therefore, using Equation (6) can be highly inefficient, as every time particle crosses an interface of different materials, one has to bring back the particle to the interface and sample the path length using the correct Σ_t .
- In such situations, one may sample the distance (r) in terms of **the number of mean-free-paths** (MFPs, λ).
- The MFP, which is **the expected (average) distance that a particle travels freely without any interaction**, is calculated by

$$\lambda = \mathbb{E}(r) = \int_0^{\infty} r \cdot \Sigma_t e^{-\Sigma_t r} dr \quad (7)$$

- Using the integration-by-parts approach, the above equation reduces to

$$\lambda = -re^{-\Sigma_t r} \Big|_0^\infty - \int_0^\infty e^{-\Sigma_t r} dr = -\frac{1}{\Sigma_t} e^{-\Sigma_t r} \Big|_0^\infty = \frac{1}{\Sigma_t} \quad (8)$$

- Therefore, Equation (6) can be re-written in terms of MFP as

$$r = -\frac{\ln \eta}{\Sigma_t} = -\frac{\ln \eta}{\frac{1}{\lambda}} = -\lambda \ln \eta \quad (9)$$

- Define the **number of MFPs** as:

$$b = \frac{r}{\lambda} = -\ln \eta \quad (10)$$

The procedure for using the number of MFP approach

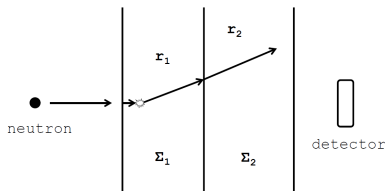
- **Step 1:** determine $b_m = \sum_{i=1}^m \Sigma_{t,i} r_i$ for each subregion (m) along the direction of motion of particle, as shown in the figure.
- **Step 2:** generate a random number (η), march from region $\hat{m}$ (particle position) toward the last subregion along the particle direction until the following inequality is satisfied, indicating that the particle is in region m .

$$b_{m-1} < -\ln \eta \leq b_m \quad (11)$$

- **Step 3:** calculate the distance r that the neutron travels **within the m^{th} region** as

$$r = -\frac{\ln \eta + b_{m-1}}{\Sigma_{t,m}} \quad (12)$$

- It is worth noting that the number of MFP approach become more beneficial with increasing number of regions, especially if $\Sigma_{t,m}$ is very small for all subregions.



Outline

2. MC for Simplified Particle Transport

2.1. Sampling path length

2.2. Sampling interaction type

2.3. Selection of scattering angle

Sampling interaction type for a binary process (absorption/scattering)

- Following its free flight, the particle undergoes an interaction. The question is: what type of interaction?
- As mentioned, we are considering only “absorption (capture)” and “elastic scattering”.
- This is a binary random process. The probability of absorption is

$$p_a = \frac{\Sigma_a}{\Sigma_t} \quad (13)$$

- The probability of scattering is:

$$p_s = \frac{\Sigma_s}{\Sigma_t} = 1 - \frac{\Sigma_a}{\Sigma_t} = 1 - p_a \quad (14)$$

- The procedure for this binary set of interactions is given below:
 - 1 Generate a random number (η);
 - 2 If $\eta \leq p_a = \frac{\Sigma_a}{\Sigma_t}$, the particle is absorbed, otherwise it is scattered.
- What if there are $N > 2$ types of interactions?

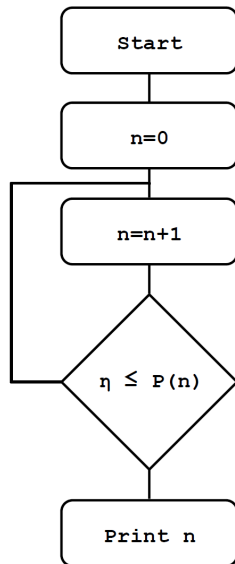
Sampling interaction type for $N > 2$ types of interactions

- **Step 1:** determine the interaction probabilities (p_n) corresponding to different interaction types, i.e., $n = 1, 2, \dots, N$.
- **Step 2:** for any arbitrary list of interaction types, determine the corresponding CDFs

$$P_n = \sum_{k=1}^n p_k \quad (15)$$

- **Step 3:** generate a random number (η), and if η satisfies the following inequality, then the outcome is interaction type n .

$$P_{n-1} < \eta \leq P_n \quad (16)$$



Outline

2. MC for Simplified Particle Transport

2.1. Sampling path length

2.2. Sampling interaction type

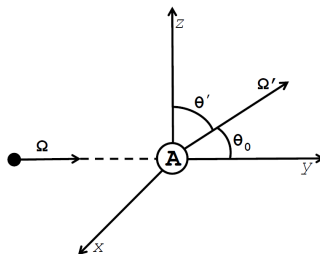
2.3. Selection of scattering angle

- If the interaction type is **scattering**, we need to determine the **scattering angle**.
- To do so, we need to derive a **PDF for scattering angle distribution**.
- From **one-speed particle transport** theory, the **differential scattering cross-section** is expressed by

$$\Sigma_s(\hat{\Omega} \rightarrow \hat{\Omega}')d\hat{\Omega}' \quad (17)$$

which is defined as the probability per unit length that a particle moving along direction $\hat{\Omega}$ will scatter into a solid angle² $d\hat{\Omega}'$ about the direction $\hat{\Omega}'$. Here the unit of the differential scattering cross-section is $\frac{1}{\text{cm} \cdot \text{steradian}}$.

- The figure on the right depicts definition of the differential scattering formulation in a 3-D (x, y, z) coordinate system.



²In Equation (1), as well as Lecture 5a, we defined that the scattering is from direction $\hat{\Omega}$ to direction $\hat{\Omega}'$ within $d\hat{\Omega}$. Here the notations have changed in order to be consistent with the book by Haghghat.

- Considering that in the one-speed transport theory, the differential scattering is dependent (only) on the scattering angle, i.e., $\mu_0 = \cos \theta_0 = \hat{\Omega} \cdot \hat{\Omega}'$ (θ_0 is the angle between $\hat{\Omega}$ and $\hat{\Omega}'$), therefore the differential scattering formulation reduces to

$$\Sigma_s(\hat{\Omega} \rightarrow \hat{\Omega}') = \Sigma_s(\mu_0) \quad (18)$$

- The solid angle is expressed by

$$d\hat{\Omega}' = d\mu' d\phi', \quad \text{for } -1 \leq \mu' \leq 1, \quad 0 \leq \phi' \leq 2\pi \quad (19)$$

where $\mu' = \mathbf{i} \cdot \hat{\Omega}' = \cos(\theta')$, and θ' and ϕ' are referred to as **polar angle** and **azimuthal angle**³, respectively. Here, $\mathbf{i}$ is the unit vector along the x -axis in a 3-D frame of reference.

³Here, ϕ is not be to confused with the scalar neutron flux $\phi(x)$ in Equation (2). Also note that in Lecture 5a, we used ω to represent the azimuthal angle.

- Since the differential scattering process is comprised of two RVs, i.e., **scattering interaction and direction after scattering (scattering angle)**, then we have

$$\Sigma_s(\hat{\Omega} \rightarrow \hat{\Omega}')d\hat{\Omega}' = \Sigma_s(\mu_0)d\hat{\Omega}' = \Sigma_s(\mu_0)d\mu'd\phi' = \Sigma_s p'(\mu_0)d\mu'd\phi' \quad (20)$$

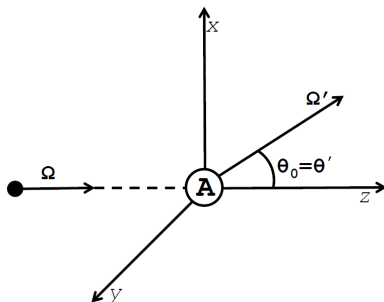
where Σ_s is the scattering cross-section (i.e., probability per unit length for scattering) and $p'(\mu_0)$ is the probability per steradian that the scattered particle will change direction to new direction $\hat{\Omega}'$ within the solid angle $d\hat{\Omega}'$.

- To proceed further, it is necessary to find a relation between μ' and μ_0 .

- To avoid mathematical complication, here we first **perform a rotation of system of coordinates** such that z -axis become aligned with the particle direction before scattering ($\hat{\Omega}$).
- Then, as shown in the figure below, $\mu' = \mu_0$ and $\phi' = \phi_0$, and Equation (20) becomes

$$\Sigma_s(\mu_0)d\mu_0d\phi_0 = \Sigma_s p'(\mu_0)d\mu_0d\phi_0 \quad (21)$$

- Note that normally after a rotation of the system of coordinates, it is necessary to modify equations to account for rotation effects, but this is not necessary here as the variable of interest is μ_0 that does not change. Therefore, $\Sigma_s(\mu_0)$ and $p'(\mu_0)$ are not affected.



- Next, we integrate Equation (21) over $\phi_0 \in [0, 2\pi]$, and solve for $p(\mu_0)$ as follows

$$p(\mu_0)d\mu_0 = \frac{\int_0^{2\pi} \Sigma_s(\mu_0)d\phi_0 d\mu_0}{\Sigma_s} = 2\pi \frac{\Sigma_s(\mu_0)}{\Sigma_s} d\mu_0 \quad (22)$$

- Note that we have chosen to set

$$p(\mu_0) = \int_0^{2\pi} p'(\mu_0)d\phi_0 \quad (23)$$

This measure maintains that the unit of differential cross-section remains per unit steradian, as the available data commonly are given per unit steradian.

- Now, if we assume that the scattering process is **isotropic**, i.e., $\Sigma_s(\mu_0)$ is equal to a constant c , then we have to derive the value of c by considering that

$$\int_0^{2\pi} d\phi_0 \int_{-1}^{+1} d\mu_0 \Sigma_s(\mu_0) = \Sigma_s \quad \rightarrow \quad 2\pi \int_{-1}^{+1} c d\mu_0 = \Sigma_s \quad \rightarrow \quad c = \frac{\Sigma_s}{4\pi} \quad (24)$$

- Therefore, **the differential scattering formulation for an isotropic scattering** is⁴

$$p(\mu_0) = \frac{1}{2} \quad (25)$$

- Then we can derive the FPMC for sampling the scattering angle (μ_0) as follows

$$\int_{-1}^{\mu_0} \frac{1}{2} dt = \eta_1 \quad (26)$$

- To sample an isotropic scattering process:**

$$\mu_0 = 2\eta_1 - 1 \quad (27)$$

⁴Different from the Equations (6.21) and (6.22) in the textbook.

- Now, we have to **sample μ' , cosine of polar angle for direction $\hat{\Omega}'$ after scattering**. To so, we utilize the following identity⁵:

$$\mu' = \mu\mu_0 + \sqrt{1 - \mu^2} \cdot \sqrt{1 - \mu_0^2} \cdot \cos(\phi_0) \quad (28)$$

- Given that ϕ_0 follows a uniform distribution in $[0, 2\pi]$, $p(\phi_0) = \frac{1}{2\pi}$.
- The corresponding FFMC is expressed by

$$P(\phi_0) = \int_0^{\phi_0} \frac{1}{2\pi} d\phi_0^* = \eta_2 \quad (29)$$

- To sample the azimuthal angle:

$$\phi_0 = 2\pi\eta_2 \quad (30)$$

⁵In the textbook, Equation (6.24), $\mu' = \mu\mu_0 + \sqrt{1 - \mu^2} \cdot \sqrt{1 - \mu'^2} \cdot \cos(\phi_0)$ was used. However, there should be no μ' on the right hand side.

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- Based on the discussions in the previous section, a flowchart for MC particle transport in a 1-D shield, considering one-speed theory, scattering and capture interactions, and isotropic scattering, is presented in [MC5b_MC_algorithm_for_a_1-D_shield.pdf](#).
- In the flowchart, **the particle weight w has been set to 1**. It will be discussed in future lectures.
 - 1 First the problem is initialized by defining parameters, including the maximum number of particles ($n_{\max}$), shield thickness (L), relative uncertainty tolerance (**tol**), Σ_t and Σ_s .
 - 2 Next, the particle counter is incremented and the new particle's location x , direction μ , and weight w are initialized.
 - 3 Then, path-length (r) is determined using Equation (6), and from it the particle position x is calculated.
 - 4 The particle position is compared to the shield boundaries to maintain that it is inside the shield. If true, then, interaction type is sampled. If the particle scatters, a new direction is obtained, followed by looping back to sample a new path-length.
- This process continues until a particle is absorbed or exits the shield. Then the particle weight is added to the appropriate counter: **absorption (abs)**, **transmission (trn)**, or **reflection (ref)**.
- If the counter is **less than maximum number of particles** and the estimated relative uncertainty R is **greater than the tolerance**, a new particle is initialized.

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- The results of a MC simulation should include **average quantities** and their **associated uncertainties**. Further, a MC simulation generally is terminated after a prescribed **precision** is achieved, i.e., **when the relative uncertainty is less than a given tolerance**.
- Considering that the number of particle histories, N , is very large, the variance corresponding to the sample average is calculated by applying the Central Limit Theorem (CLT):

$$\sigma_{\bar{x}}^2 = \frac{\sigma_x^2}{N} \quad (31)$$

- Since it is not possible to guess if N is sufficiently large for a given problem, one should at least examine the behavior of relative uncertainty and the figure of merit (FOM). The FOM is:

$$\text{FOM} = \frac{1}{R_{\bar{x}}^2 \cdot T} \quad (32)$$

where $R_{\bar{x}}^2$ is the estimated relative uncertainty ($R_{\bar{x}} = \frac{\sigma_{\bar{x}}}{\bar{x}}$), and T is the computation time.

- To examine the relative uncertainty, we determine the ratio of relative uncertainties (R_1 and R_2) for increasing number of histories, from N_1 to N_2 as follows

$$\frac{R_2}{R_1} = \frac{\frac{\sigma_{\bar{x},2}}{\bar{x}_2}}{\frac{\sigma_{\bar{x},1}}{\bar{x}_1}} = \frac{\frac{\sigma_{x,2}}{\bar{x}_2 \sqrt{N_2}}}{\frac{\sigma_{x,1}}{\bar{x}_1 \sqrt{N_1}}} \quad (33)$$

- If CLT is valid, then $\frac{\sigma_{x,2}}{\bar{x}_2} \approx \frac{\sigma_{x,1}}{\bar{x}_1}$. Therefore $\frac{R_2}{R_1}$ reduces to

$$\frac{R_2}{R_1} = \sqrt{\frac{N_1}{N_2}} \quad (34)$$

Hence, a reliable result of a MC simulation should follow Equation (34).

- For the FOM, if CLT is valid, then

$$R_{\bar{x}} \propto \frac{\sigma_x}{\sqrt{N}} \cong \frac{C_1}{\sqrt{N}} \quad (35)$$

and considering computer time (T) should change with the number of histories, i.e., $T \cong C_2 N$, then the FOM reduces to

$$\text{FOM} \propto \frac{1}{\left(\frac{C_1}{\sqrt{N}}\right)^2 \cdot (C_2 N)} \cong C \quad (36)$$

- This means that for a reliable MC simulation, it is necessary that the corresponding FOM fluctuates about a constant value.